

Exercise 2: Factory Method Pattern

Document.java

```
public interface Document{  
    void open();  
}
```

ReportFile.java

```
public class ReportFile implements Document{  
    public void open(){  
        System.out.println("Report Document Opened");  
    }  
}
```

InvoiceFile.java

```
public class InvoiceFile implements Document{  
    public void open(){  
        System.out.println("Invoice Document Opened");  
    }  
}
```

DocumentCreator.java

```
public abstract class DocumentCreator{  
    public abstract Document createDocument();  
}
```

ReportCreator.java

```
public class ReportCreator extends DocumentCreator{  
    public Document createDocument(){  
        return new ReportFile();  
    }  
}
```

InvoiceCreator.java

```
public class InvoiceCreator extends DocumentCreator{
    public Document createDocument(){
        return new InvoiceFile();
    }
}
```

FactoryDemo.java

```
public class FactoryDemo
{
    public static void main(String[] args)
    {
        DocumentCreator creator1 = new ReportCreator();

        Document doc1 = creator1.createDocument();

        doc1.open();

        DocumentCreator creator2 = new InvoiceCreator();

        Document doc2 = creator2.createDocument();

        doc2.open();
    }
}
```

Output:

```
Report Document Opened
Invoice Document Opened

...Program finished with exit code 0
Press ENTER to exit console.█
```